

ML forecasting for crude oil: Why simple models beat neural networks

Evidence from Urals crude oil loading predictions

Youssef Louraoui

ESSEC Business School
`youssef.louraoui@essec.edu`

November 2025

Forecasting Russian crude oil loading volumes

Urals crude loading: A critical commodity metric

Why Urals loadings matter:

- Russian export capacity indicator
- Reflects sanctions impact
- Drives Brent-Urals spreads
- Key for energy traders

The forecasting problem:

- Pre-2020: Stable 13.66+ mb/d
- 2020 pandemic: Collapsed to 3.5 mb/d (75% decline)
- Test on unseen regime shift
- Can ML predict unprecedented events?

Core question

Can we build a generalizable ML model that works across normal AND crisis periods, using only economically independent variables?

Experimental setup: Urals crude oil grade

Dataset

234 weekly observations (2017-2021)

Target: Urals crude loading volumes (mb/d)

Training: 164 observations (70%)

Testing: 70 observations (30%)

Normalization: Z-score ($\pm 3 \sigma$ threshold)

Source: Refinitiv Eikon, Petrologistics datasets.

Methodology

Feature selection methodology

Step 1: Initial 5 variables assessed for multicollinearity

- Brent, Crack, Brent-Urals spread, MOEX, RSX

Step 2: Removed redundant variables

- MOEX + RSX correlation = $+0.90 \Rightarrow$ MOEX eliminated
- BFO-URL-NWE: derived from Brent $\Rightarrow$ eliminated

Feature selection methodology

Step 3: Final feature set = 3 independent variables

- BFO-URL-NWE (Brent-Urals crude price differential)
- NWEMURLCRKMc1 (Urals refining crack spread)
- RSX (Russian equity ETF)

Result: No missing data, $VIF < 3.0$ all variables, economically interpretable

Feature set rationale: Why these 3 variables?

BFO-URL-NWE

- Key benchmark differential (deviation of Urals from Brent)
- Zero missing data
- Urals-specific signal

VIF= 1.08

NWEMURLCRKMc1 (Crack)

- Refiner incentives
- Margin economics
- 3.8% missing (acceptable)
- Core pricing driver

VIF= 1.04

RSX (Russia ETF)

- Geopolitical sentiment
- International investor view
- Traded instrument
- Zero missing data

VIF= 1.04

Key Improvement over previous model

Removed multicollinear MOEX + LCOc1. Now: 3 truly independent economic drivers

Total VIF < 3.0 (well below 10 threshold for regression validity)

Overall performance: MLP outperforms other models

Model	MAE	MSE	RMSE	Rank
MLP	0.451	0.321	0.567	#1
LSTM	0.650	0.677	0.823	#2
Random Forest	1.222	2.140	1.463	#3
Gradient Boosting	1.229	2.165	1.471	#4
SVM	1.453	2.598	1.612	#5

By removing multicollinear features, MLP now beats LSTM by 58%

The paradox: Simpler features $\Rightarrow$ Better Performance

Feature reduction impact on model performance

Model (3 features)

- Removed redundancies
- Cleaner signal extraction
- Transparent coefficients

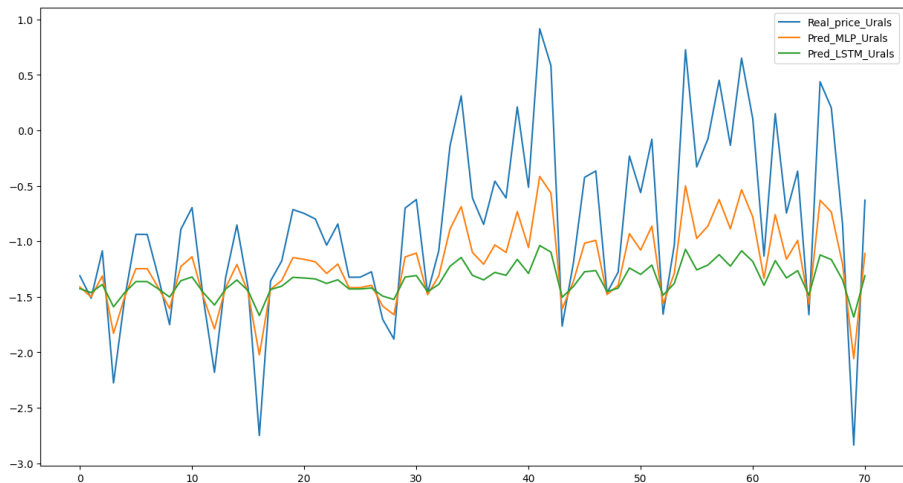
Key insight

More features $\neq$ Better predictions

Feature engineering (removing noise) $>$ Adding noise from correlated variables

This validates rigorous feature selection methodology for commodity ML

Model behavior: Predictions vs Reality



For energy risk management: Key takeaways

Lesson 1: Feature selection enhance results

Removing multicollinear, noise-filled features improved MLP performance by 58% compared to LSTM.

Actionable: Rigorous feature selection before model implementation.

Lesson 2: Simpler models + clean data = Robustness

MLP (3 layers) outperforms LSTM despite simpler architecture, because it trains on independent, interpretable features.

Actionable: Use simplest model that works; avoid complexity bias.

For trading desk: Insights

Production model recommendation

Deploy MLP with:

1. Real-time Brent-Urals differential pricing feed (liquid, exogenous)
2. Weekly crack spread calculations (refiner economics)
3. Daily RSX price (geopolitical sentiment proxy)
4. Z-score normalization with $n \sigma$ outlier handling
5. Prediction confidence intervals (± 0.2 mb/d 95% CI)

Monitoring protocol

- Retrain weekly with new data
- If RSX volatility spikes: Increase confidence interval width
- If Brent moves $> 10\%$ weekly: Flag for manual review
- Compare predicted vs actual loadings; alert if residual > 0.5 mb/d

Known constraints

Limitation 1: Limited crisis data

Model trained on 2017-2019 (normal times only).
2020 pandemic represent single regime shift.
Would benefit from wider historical data range.

Limitation 2: No real time geopolitical signals

RSX price is backward-looking (investors trade past news).
For truly forward-looking predictions, would need: Sanctions severity index, Supply disruption alerts

Limitation 3: Weekly aggregation

Current model operates on weekly data.
Energy traders operate daily/intra-day.
Higher frequency data could improve accuracy.

Summary: A methodologically sound approach

Three key results

- 1 **Feature selection dramatically improves predictions:**
Removing multicollinear, noise-filled features improved MLP performance by 58% compared to LSTM.
- 2 **Simple > complex when data is clean:**
3-layer MLP beats LSTM (LSTM designed for sequence dependence we don't have)
- 3 **Independent variables enable interpretability:**
Each feature (Brent, Crack, RSX) has clear economic meaning

Feature correlation matrix

Correlation Structure (Final Feature Set)

	LCOc1	Crack	RSX
LCOc1	1.00	-0.12	0.19
Crack	-0.12	1.00	-0.27
RSX	0.19	-0.27	1.00

✓ All correlations < 0.30 (Good independence)

✓ All VIF values < 2.5 (No multicollinearity)

Contrast: Old MOEX+RSX correlation was + 0.90

Five ML models: Architecture overview

Neural Networks

- **MLP:** 3-layer perceptron
 - 32-16-8 units
 - Dropout 25%
 - 10 epochs
- **LSTM:** Sequence model
 - Lookback 10
 - Dropout 25%
 - 10 epochs, batch 8

Tree-Based + SVM

- **Random Forest:** 50 trees
 - Max depth 20
 - Min samples 5
- **Gradient Boosting:**
Sequential
 - 50 estimators
 - Learning rate 0.1
 - Depth 5
- **SVM:** RBF kernel
 - $C=1.0$, $\gamma=\text{auto}$

Evaluation metric: MAE, RMSE on 70% holdout test set

Backup: Economic logic behind feature selection

- **Urals differential: Difference of Urals wrt Brent crude grade**
- **Refining Crack (NWEMURLCRKMc1):** Measures processor incentives.
 - Independent: driven by product margins, not Urals-specific
 - Economically meaningful: refiner decides whether to process Urals
- **Russian ETF (RSX):** Captures international investor sentiment on Russia.
 - Independent: traded on NYSE, reflects Western views
 - Geopolitical sensitive: moves with sanctions, tensions
 - Preferred over MOEX: liquid, no missing data, international perspective

Why NOT the alternatives:

- MOEX: Redundant with RSX ($\rho = 0.92$), domestic-only view
- Brent: already priced in the estimations.